

ADVANCED CALCULUS & DIFFERENTIAL EQUATIONS

Total Maximum Marks: 100 | Total Questions: 11 | Time Allowed: 3 Hours

GENERAL INSTRUCTIONS:

1. All questions are compulsory unless specified otherwise.
2. Candidates are required to show all intermediate mathematical steps, derivations, and reasoning.
3. Figures and visual plots correspond to the numbered problems. Graph interpretations must be clearly justified.

Question 1

[10 Marks]

Find the general solution to the differential equation $2(d^2x / dt^2) + 4(dx / dt) + 2x = 0$. Express your answer in terms of real-valued functions.

Question 2

[15 Marks]

An underdamped oscillator has $m=2\text{ kg}$, $b=3\text{ Ns/m}$, and $k=5\text{ N/m}$. Calculate the damping ratio ζ , the decay constant γ , and the damped natural frequency ω_d .

Question 3

[20 Marks]

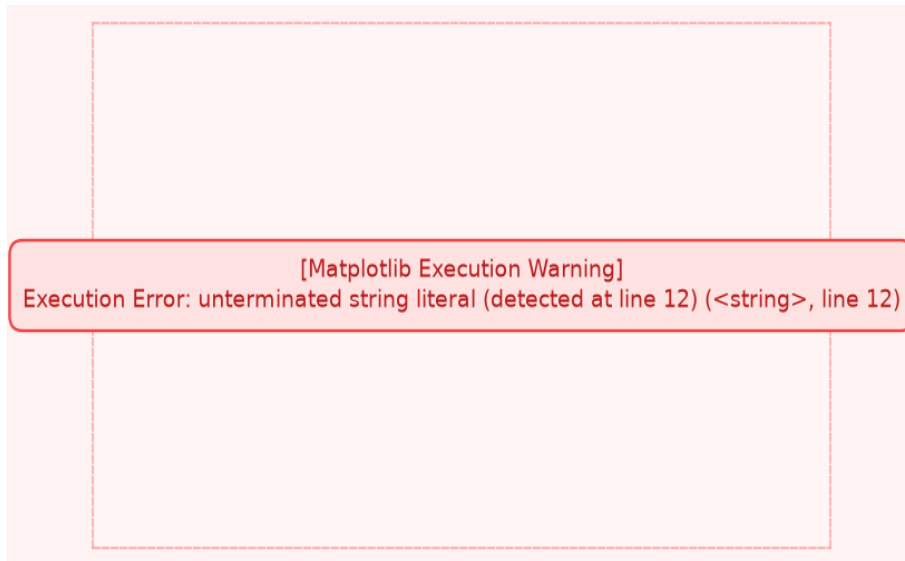
Consider the vector field $\vec{F}(x,y,z) = (2xy, x^2 + z^2, 2yz)$. (a) Show that $\vec{F}$ is conservative. (b) Find a potential function $f(x,y,z)$ such that $\nabla f = \vec{F}$. (c) Compute the line integral of $\vec{F}$ along the curve C parameterized by $\vec{r}(t) = (t, t^2, t^3)$ from $t=0$ to $t=1$.



Question 4

[20 Marks]

Verify Stokes' Theorem for the vector field $\vec{F}(x,y,z) = (y, -x, 0)$ over the surface S defined by $z = 1 - x^2 - y^2$, $x^2 + y^2 \leq 1$, oriented upward. Compute both the surface integral $\iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$ and the line integral $\oint_{\partial S} \vec{F} \cdot d\vec{r}$ and show they are equal.



Question 5

[15 Marks]

A critically damped oscillator has $m=4\text{kg}$ and $k=9\text{N/m}$. Determine the damping coefficient b required for critical damping. Solve the differential equation $4(d^2x / dt^2) + b(dx / dt) + 9x = 0$ with initial conditions $x(0) = 1\text{m}$ and $x'(0) = -2\text{m/s}$.

Question 6

[10 Marks]

Evaluate the line integral $\int_C (x^2 - y)dx + (xy + z)dy + (yz - x)dz$ where C is the path from $(0,0,0)$ to $(1,1,1)$ along the straight line $x=y=z$.

Question 7

[15 Marks]

Find the Fourier series expansion of the periodic function $f(x)$ defined on $[-\pi, \pi]$ as $f(x) = x^2$ for $-\pi \leq x < 0$ and $f(x) = x$ for $0 \leq x \leq \pi$. Compute the first three non-zero Fourier coefficients a_0 , a_1 , and b_1 .

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Question 8

[15 Marks]

Solve the initial value problem $y'' + 4y' + 4y = 0$, $y(0) = 2$, $y'(0) = -1$. Express the solution in terms of real-valued functions and sketch the graph of the solution.

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Question 9

[15 Marks]

A damped harmonic oscillator has $m=1\text{ kg}$, $b=0.5\text{ Ns/m}$, and $k=2\text{ N/m}$. Determine whether the system is underdamped, critically damped, or overdamped. If underdamped, find the general solution and compute the displacement at $t=2\text{ s}$ given $x(0)=3\text{ m}$ and $x'(0)=0\text{ m/s}$.

Question 10**[15 Marks]**

Compute the flux of the vector field $\vec{F}(x,y,z) = (x, y, z)$ through the surface S defined by $x^2 + y^2 = 1$, $0 \leq z \leq 2$, oriented outward. Use the Divergence Theorem to evaluate the flux.

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Question 11**[20 Marks]**

A periodic square wave $f(x)$ with period 2π is defined as $f(x) = 1$ for $0 < x < \pi$ and $f(x) = -1$ for $-\pi < x < 0$. (a) Find the Fourier series coefficients a_0 , a_n , and b_n . (b) Plot the first 10 terms of the Fourier series approximation to $f(x)$ on $[-\pi, \pi]$.

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